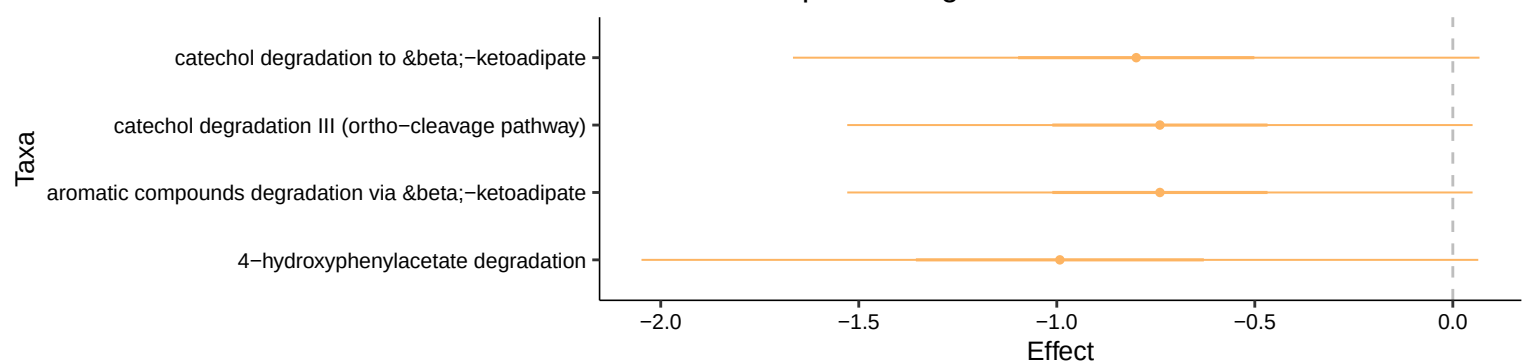
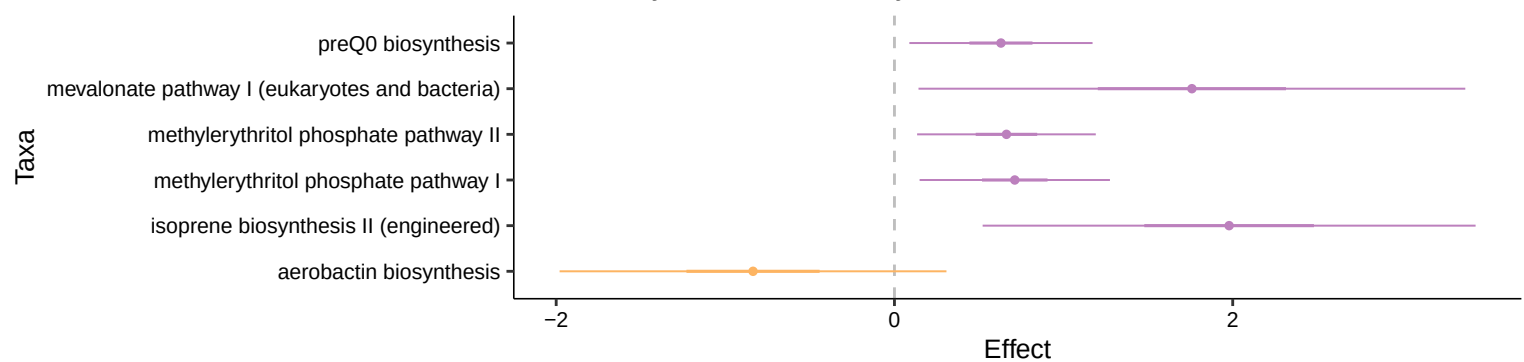


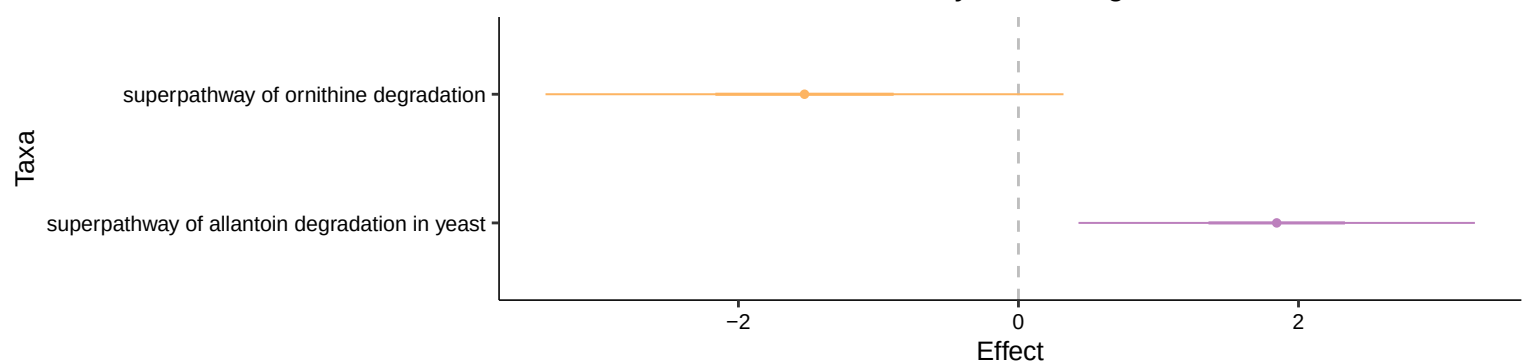
Aromatic Compound Degradation



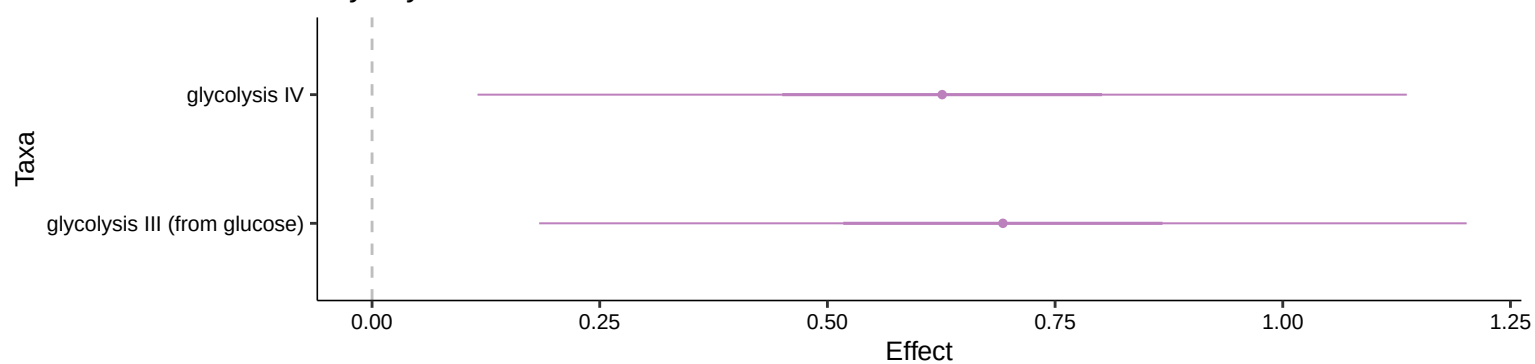
Secondary Metabolite Biosynthesis



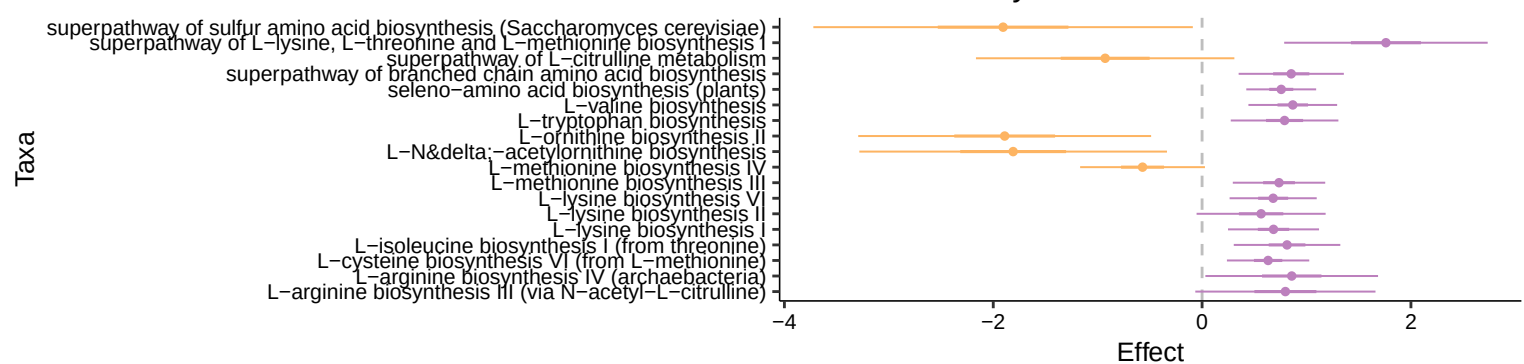
Amide, Amidine, Amine, and Polyamine Degradation



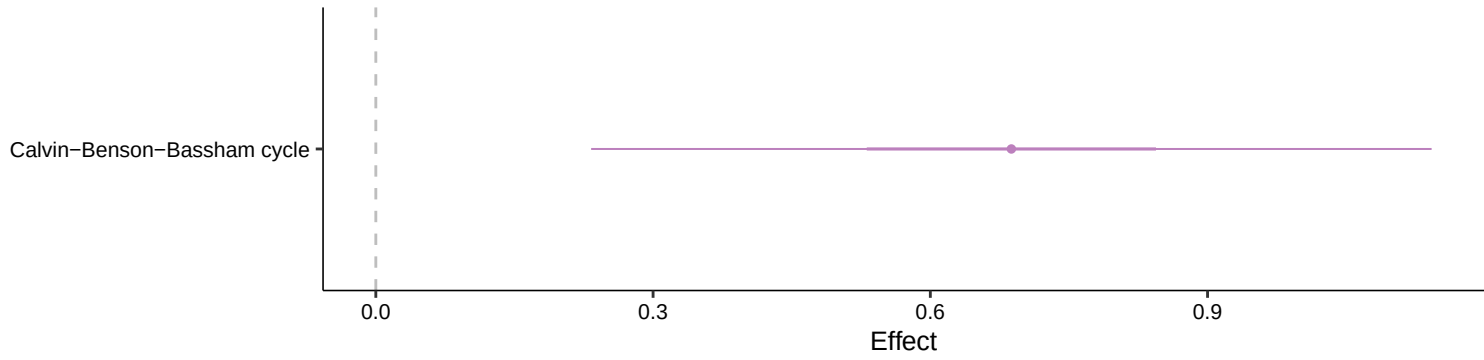
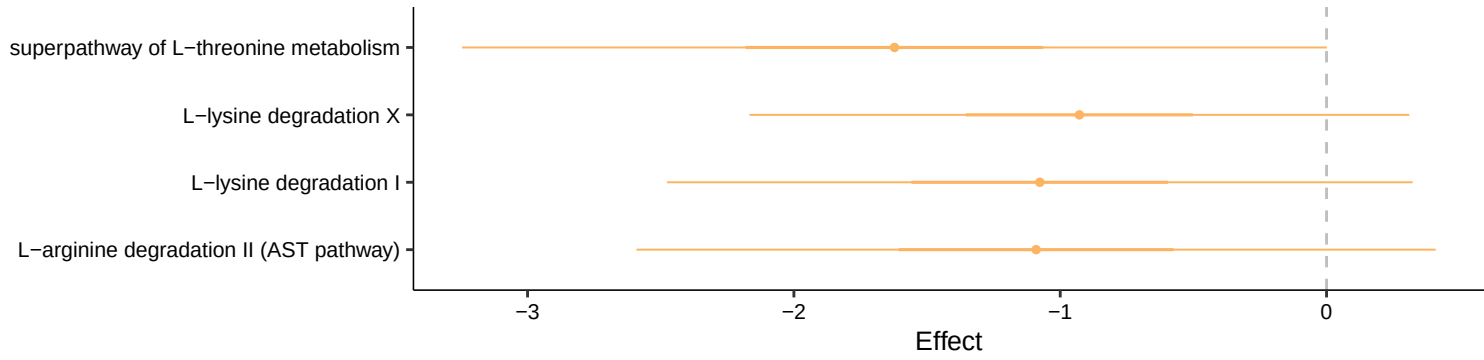
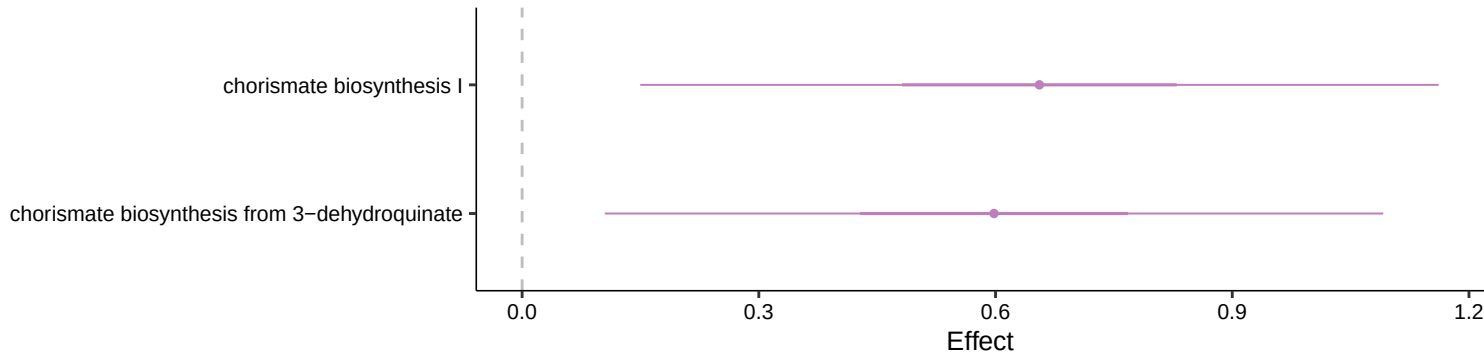
Glycolysis



Amino Acid Biosynthesis

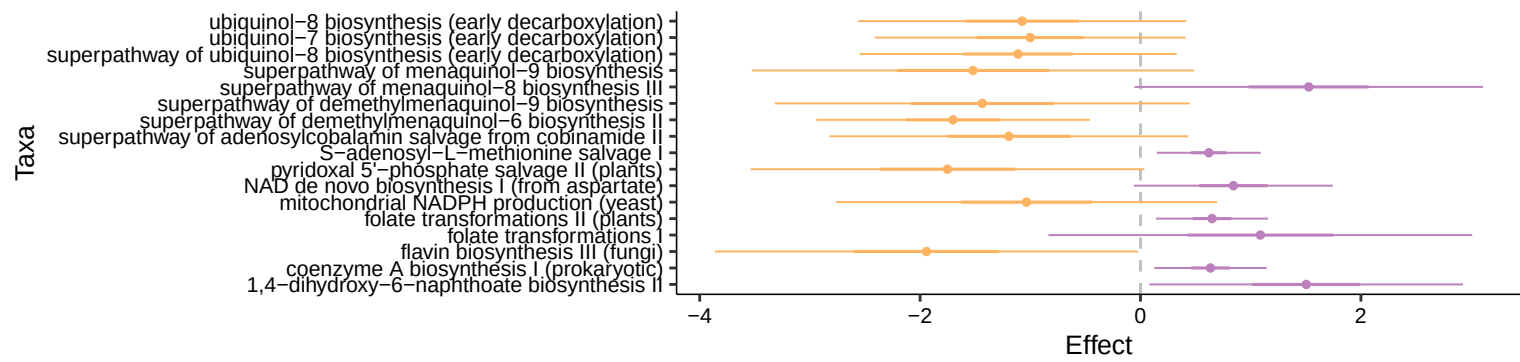


Pathway	Effect (approx.)
superpathway of sulfur amino acid biosynthesis (<i>Saccharomyces cerevisiae</i>)	-1.8
superpathway of L-lysine, L-threonine and L-methionine biosynthesis I	1.8
superpathway of L-citrulline metabolism	-1.0
superpathway of branched chain amino acid biosynthesis (plants)	0.8
seleno-amino acid biosynthesis (plants)	0.8
L-valine biosynthesis	0.8
L-tryptophan biosynthesis	0.8
L-ornithine biosynthesis II	-1.8
L-N δ -acetylornithine biosynthesis	-1.8
L-methionine biosynthesis IV	-0.5
L-methionine biosynthesis III	0.8
L-lysine biosynthesis VI	0.8
L-lysine biosynthesis II	0.8
L-lysine biosynthesis I	0.8
L-isoleucine biosynthesis I (from threonine)	0.8
L-cysteine biosynthesis VI (from L-methionine)	0.8
L-arginine biosynthesis IV (archaeobacteria)	0.8
L-arginine biosynthesis III (via N-acetyl-L-citrulline)	0.8

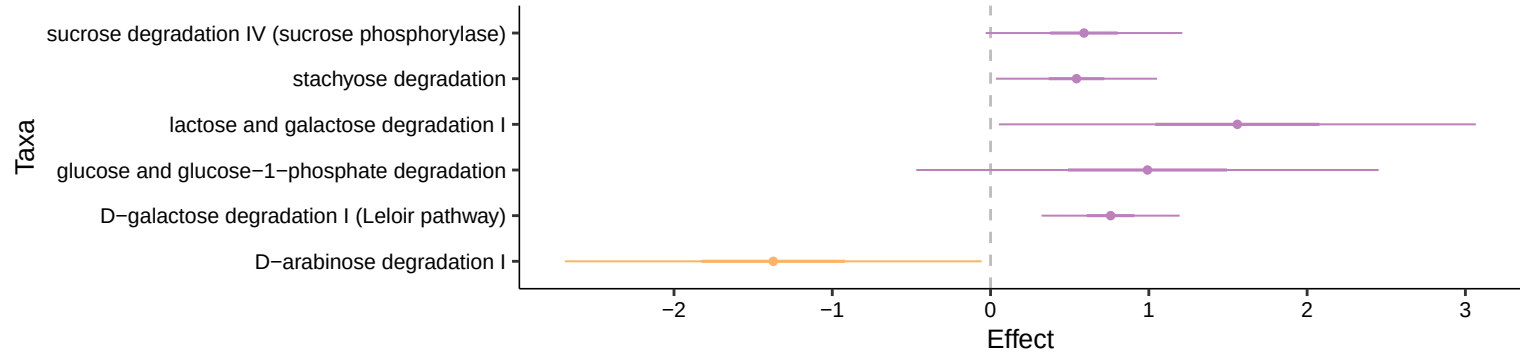


Metabolic Pathway	Effect (approx.)
ubiquinol-8 biosynthesis (early decarboxylation)	-1.0
ubiquinol-7 biosynthesis (early decarboxylation)	-1.0
superpathway of ubiquinol-8 biosynthesis (early decarboxylation)	-1.5
superpathway of menaquinol-9 biosynthesis	1.5
superpathway of menaquinol-8 biosynthesis III	-1.5
superpathway of demethylmenaquinol-9 biosynthesis	-1.5
superpathway of demethylmenaquinol-6 biosynthesis II	-1.5
superpathway of adenosylcobalamin salvage from cobinamide II	-1.0
S-adenosyl-L-methionine salvage I	0.5
pyridoxal 5'-phosphate salvage II (plants)	-1.5
NAD de novo biosynthesis I (from aspartate)	0.5
mitochondrial NADPH production (yeast)	-1.0
folate transformations II (plants)	0.5
folate transformations I	-0.5
flavin biosynthesis III (fungi)	0.5
coenzyme A biosynthesis I (prokaryotic)	0.5
1,4-dihydroxy-6-naphthoate biosynthesis II	1.5

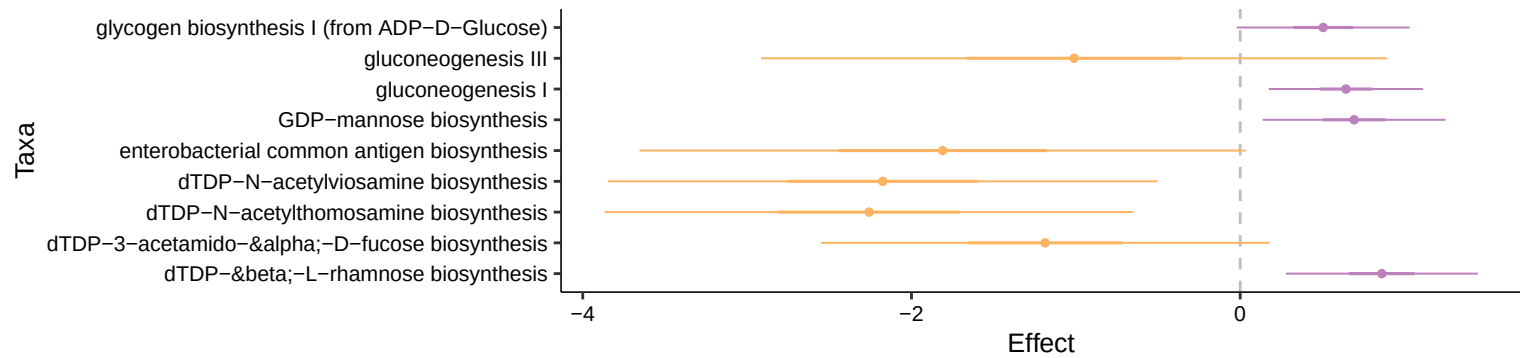
Cofactor, Carrier, and Vitamin Biosynthesis



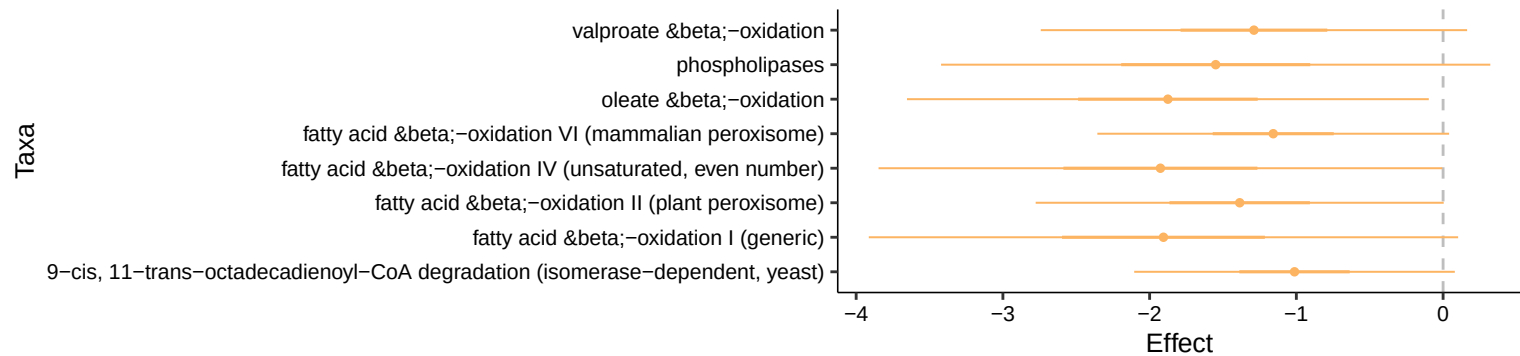
Carbohydrate Degradation



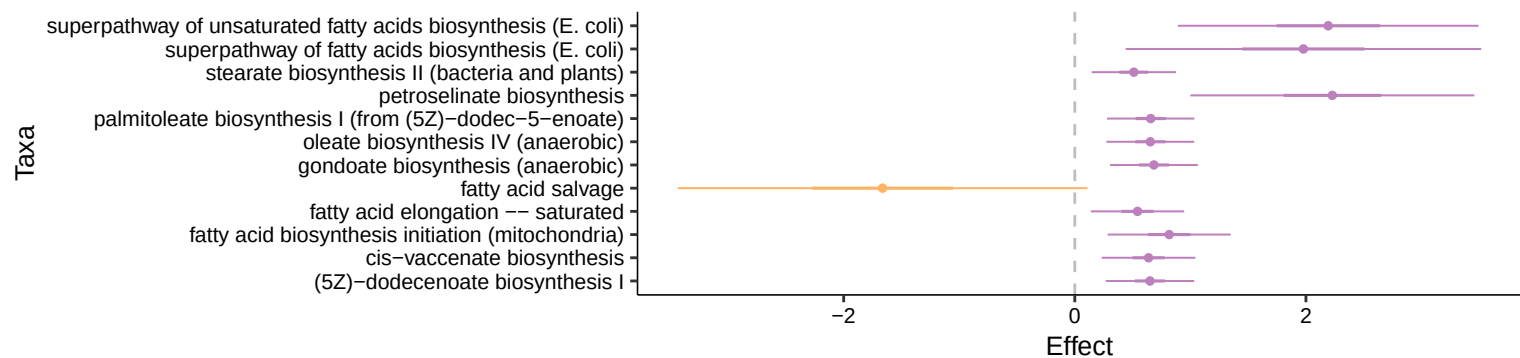
Carbohydrate Biosynthesis



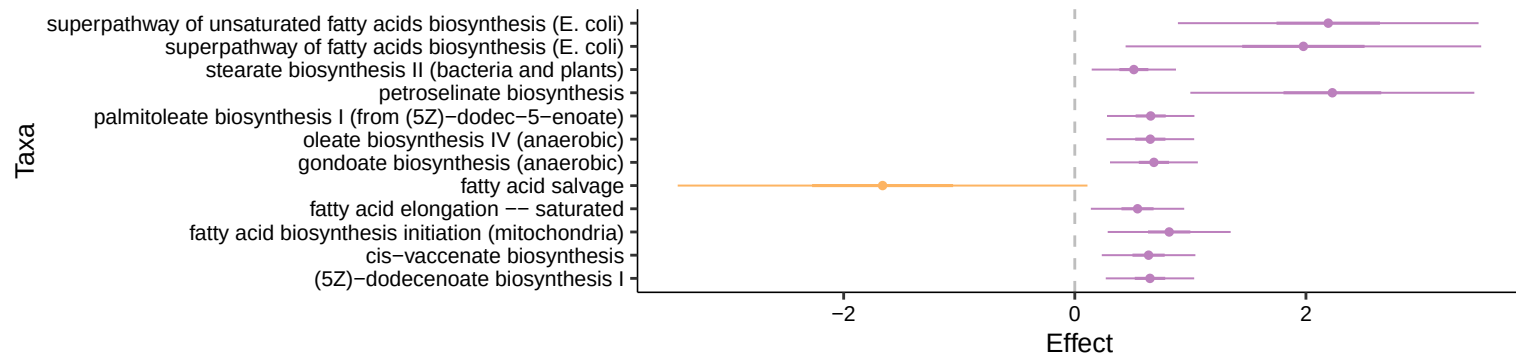
Fatty Acid and Lipid Degradation



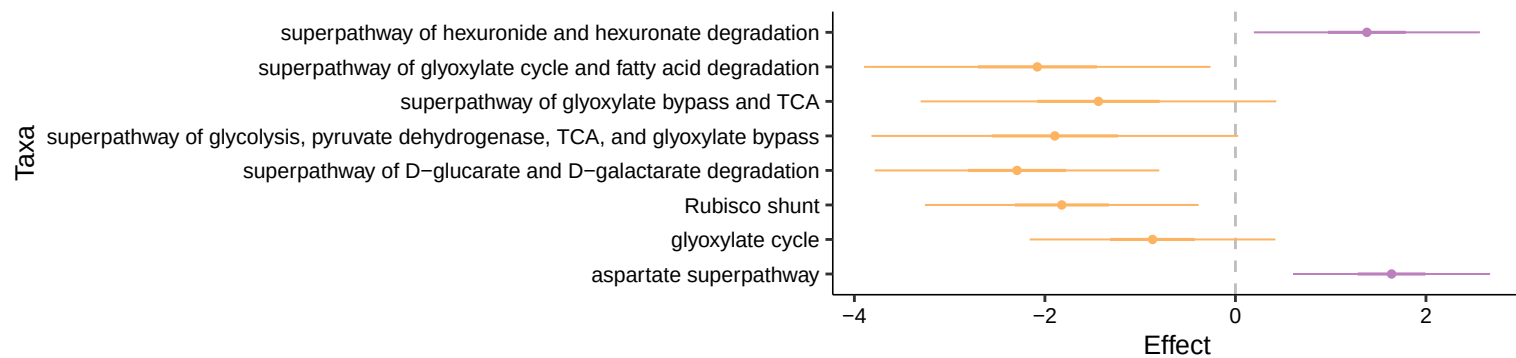
Fatty Acid and Lipid Biosynthesis



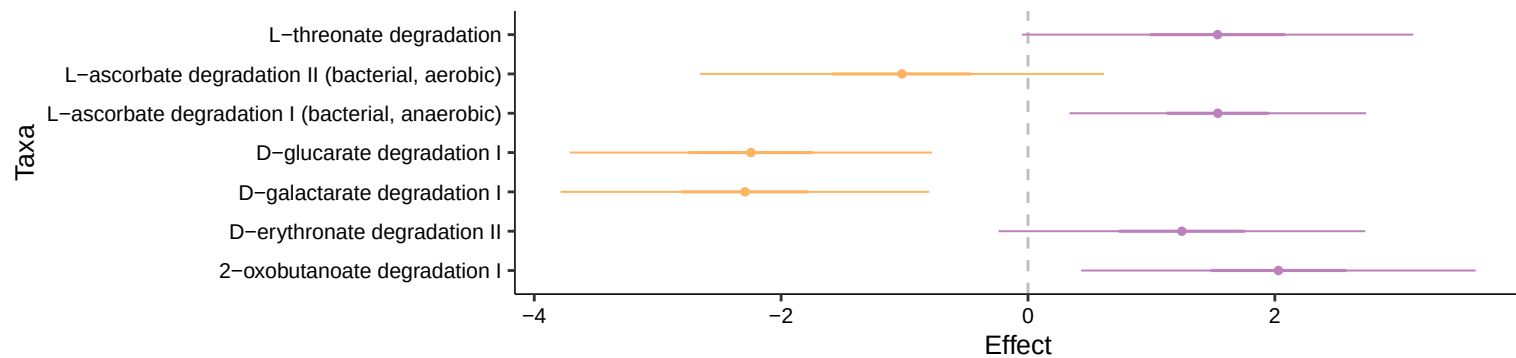
Fatty Acid and Lipid Biosynthesis



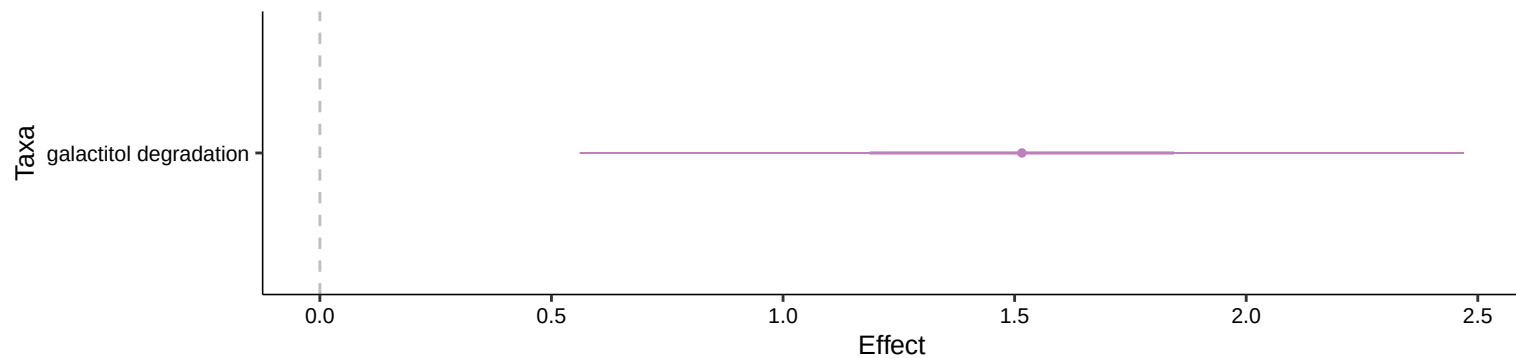
NA



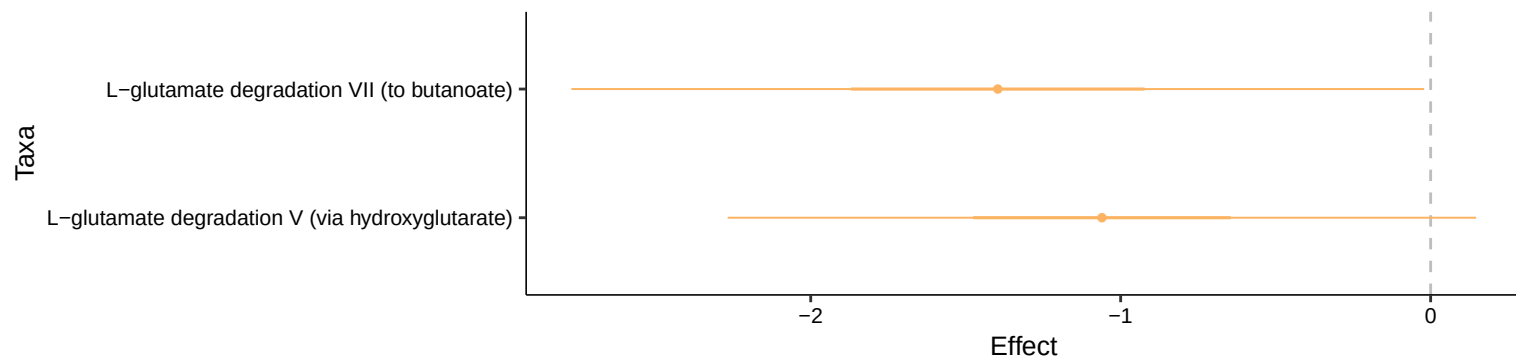
Carboxylic Acid Degradation



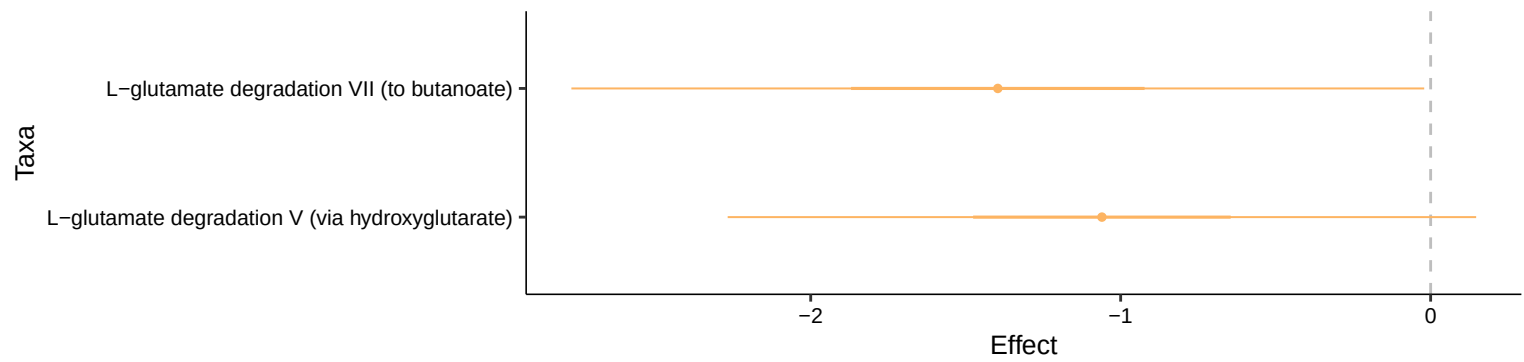
Carbohydrate Degradation&&Alcohol Degradation



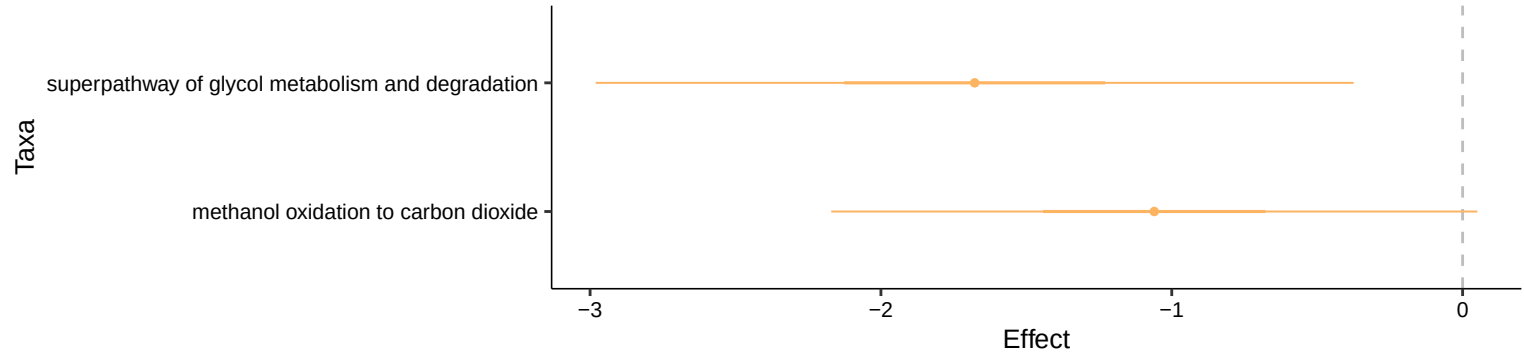
Amino Acid Degradation&&Fermentation



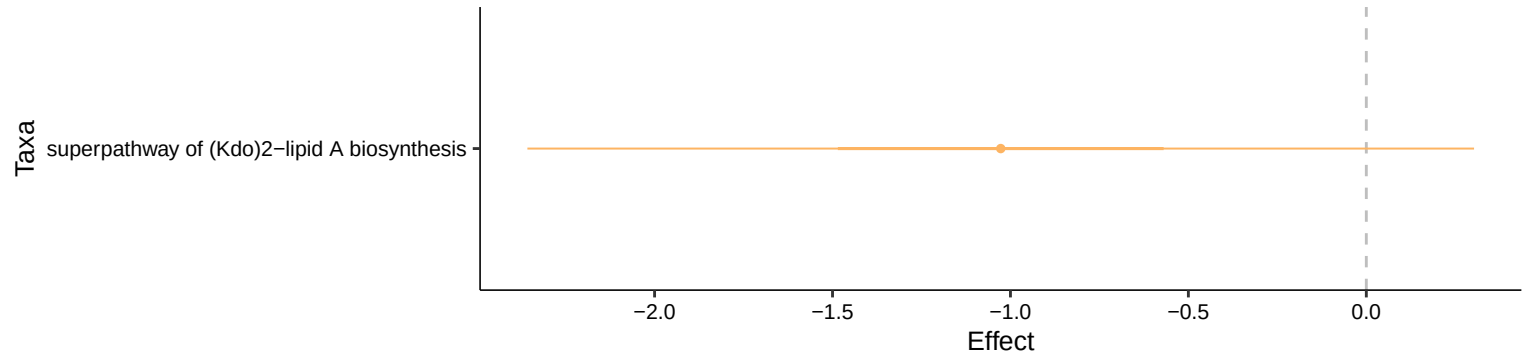
Amino Acid Degradation&&Fermentation



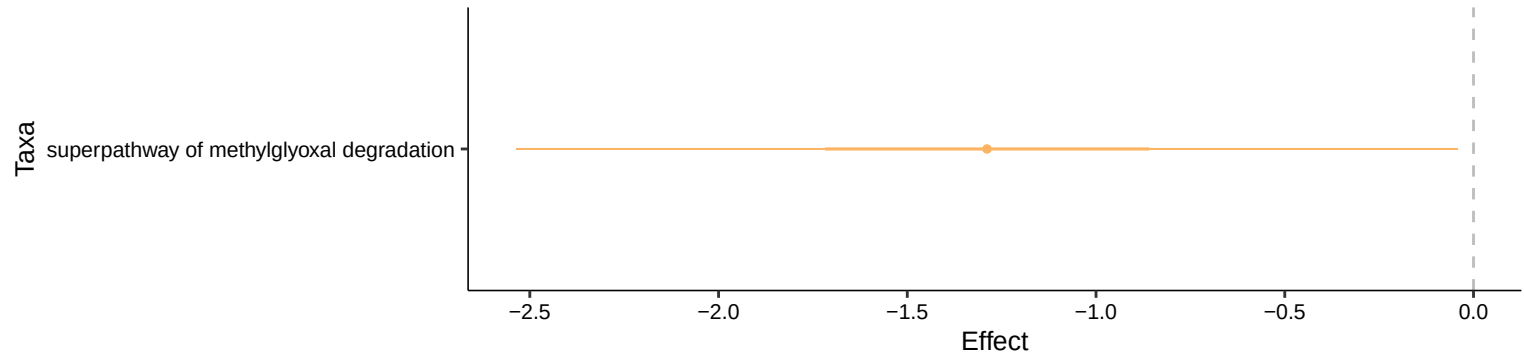
Alcohol Degradation



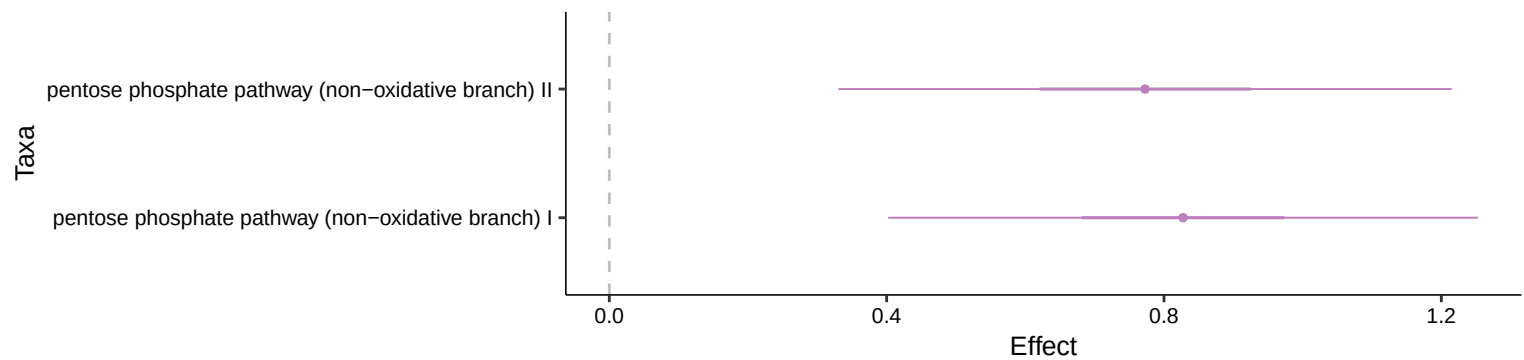
Fatty Acid and Lipid Biosynthesis&&Cell Structure Biosynthesis



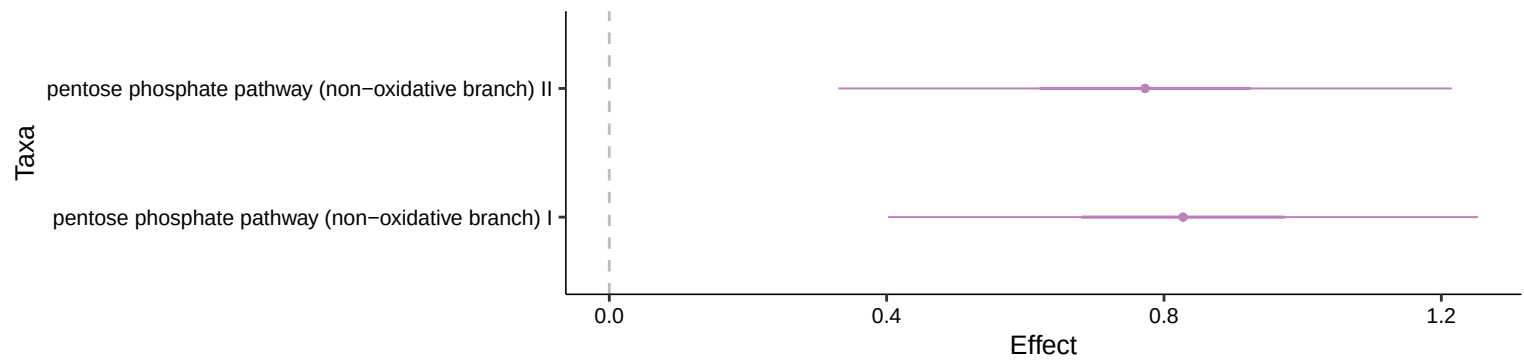
Aldehyde Degradation



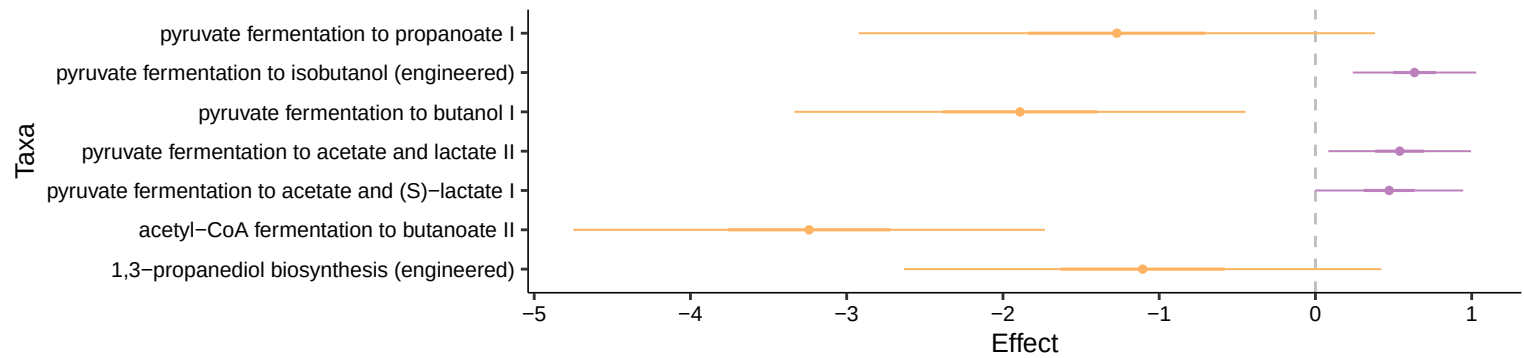
Pentose Phosphate Pathways



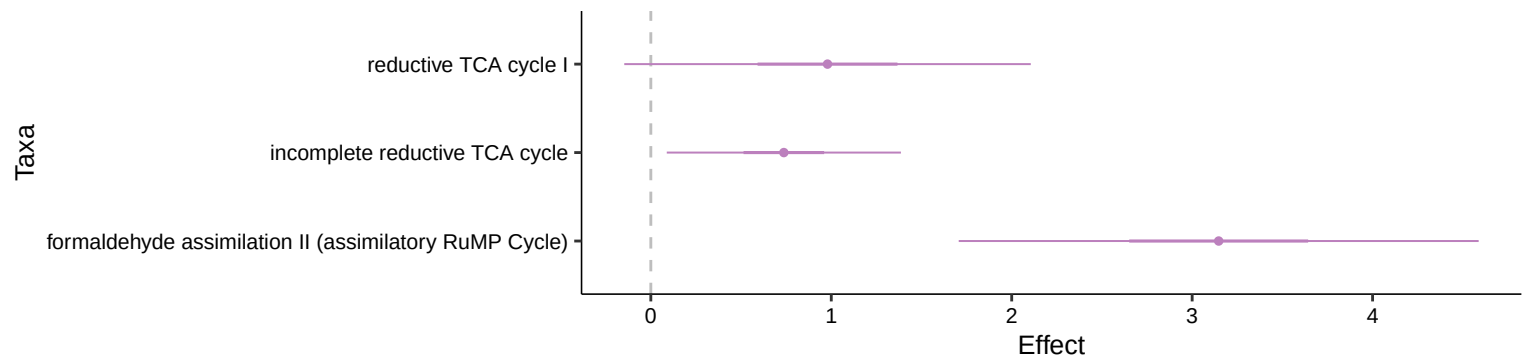
Pentose Phosphate Pathways



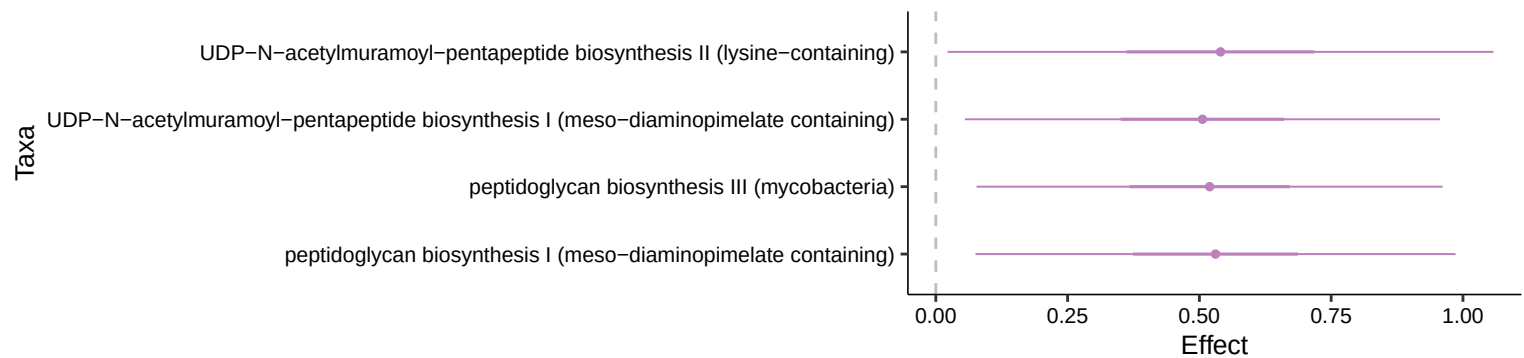
Fermentation



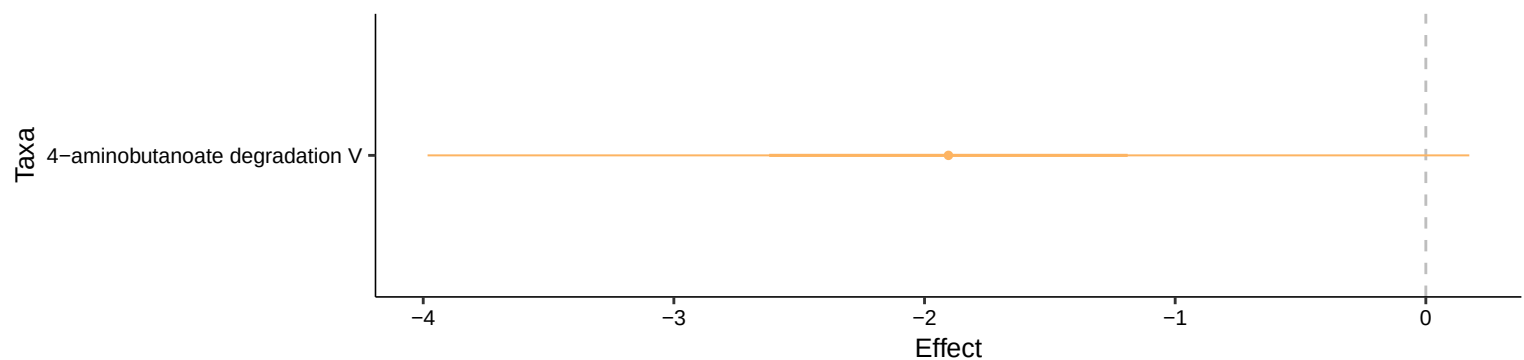
C1 Compound Utilization and Assimilation



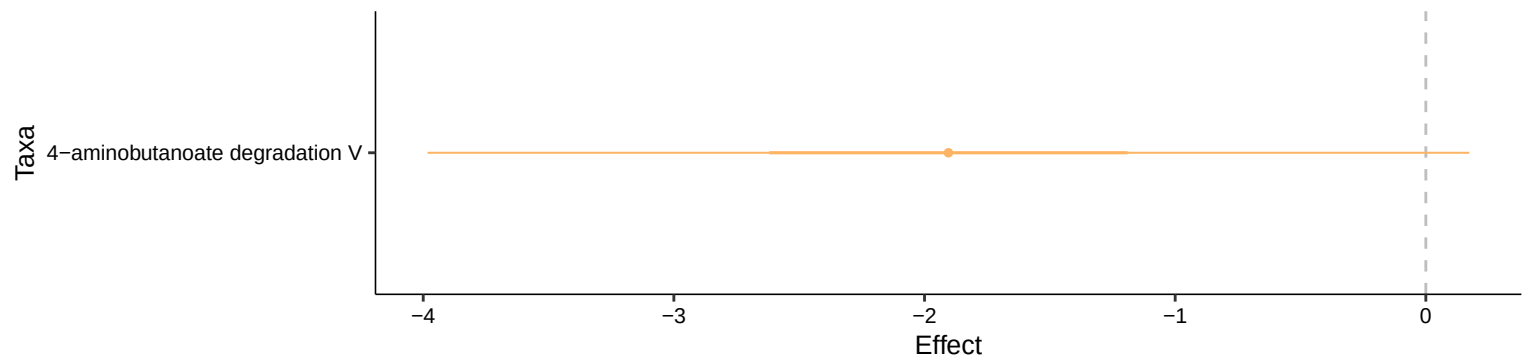
Cell Structure Biosynthesis



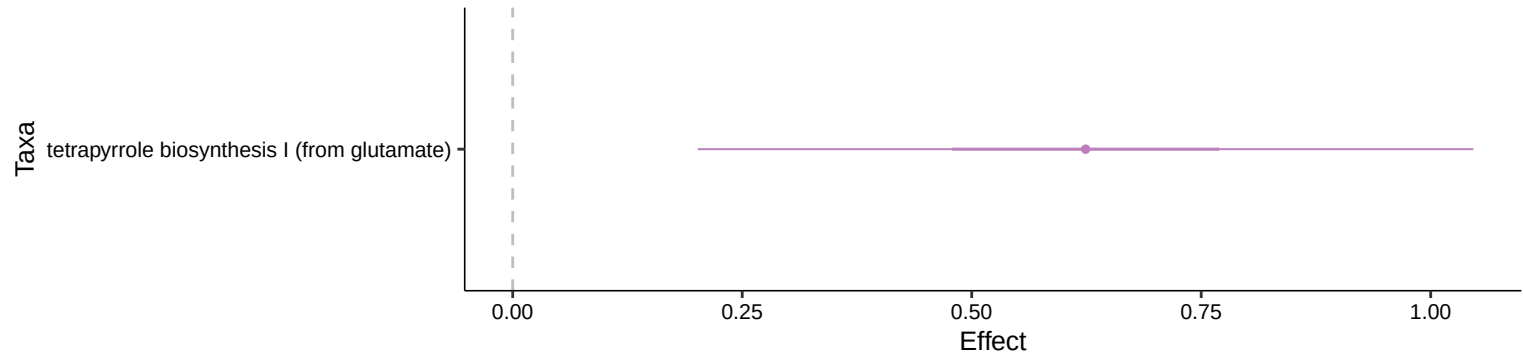
Amide, Amidine, Amine, and Polyamine Degradation&&Fermentation



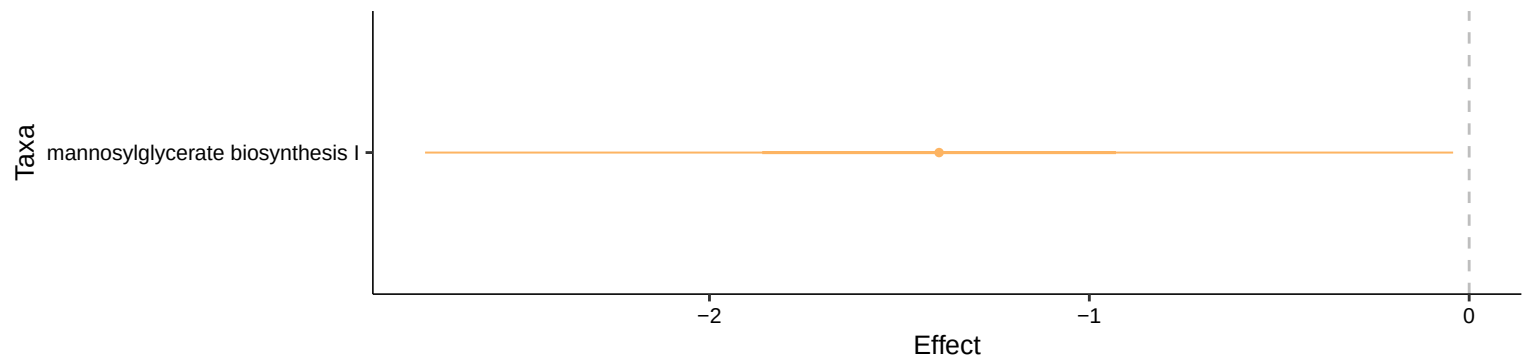
Amide, Amidine, Amine, and Polyamine Degradation&&Fermentation



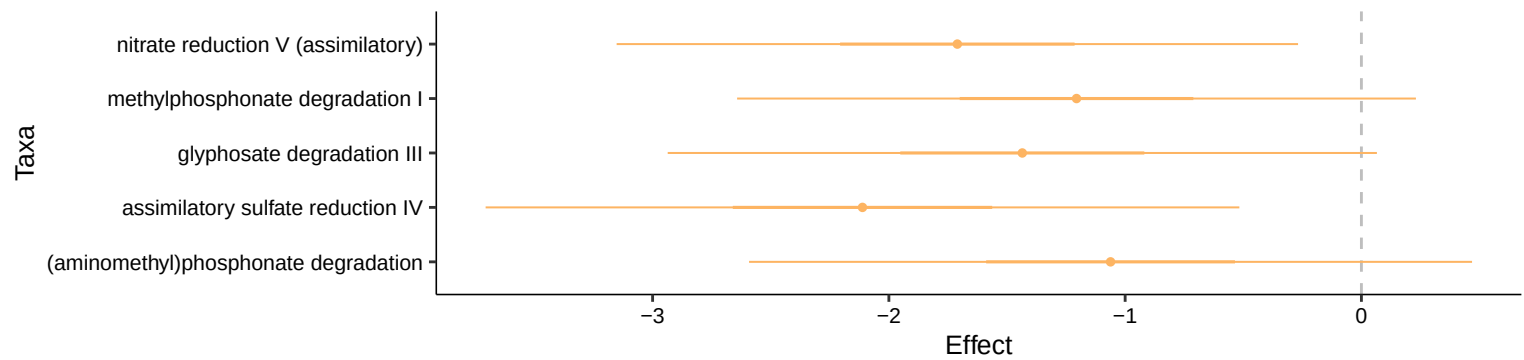
Tetrapyrrole Biosynthesis



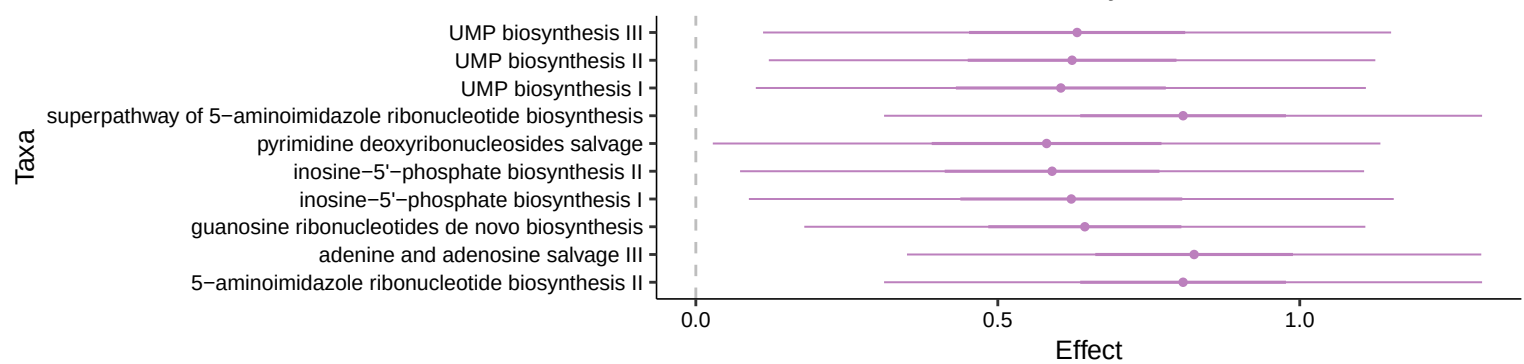
Metabolic Regulator Biosynthesis



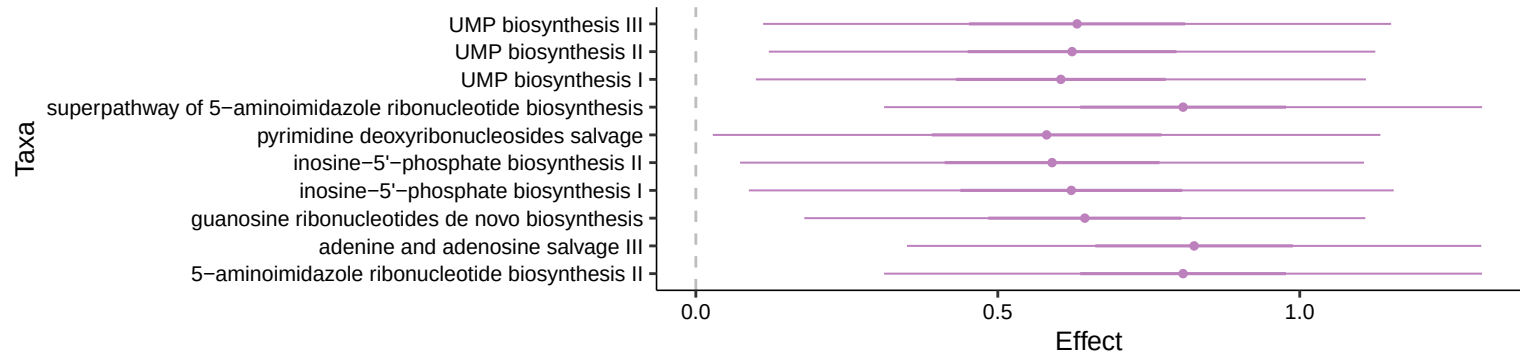
Inorganic Nutrient Metabolism



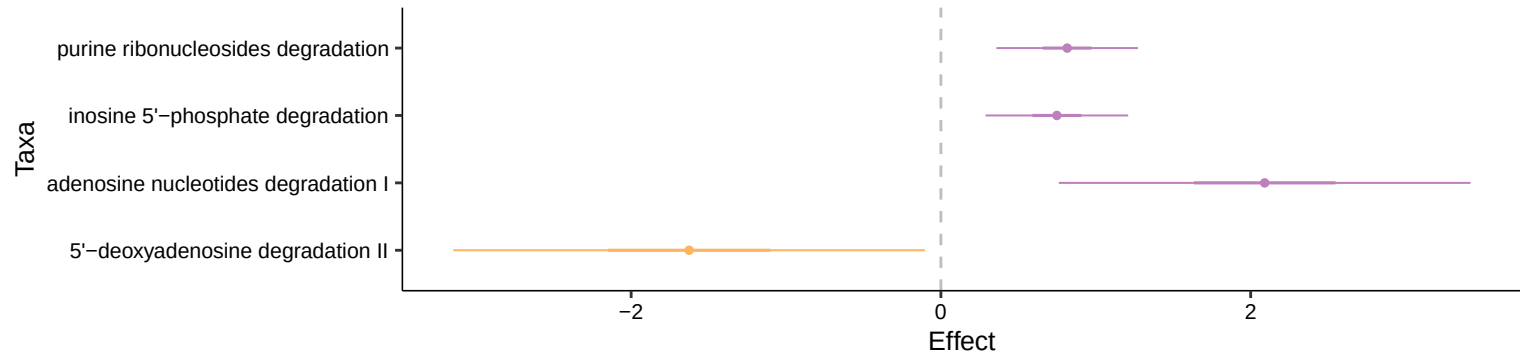
Nucleoside and Nucleotide Biosynthesis



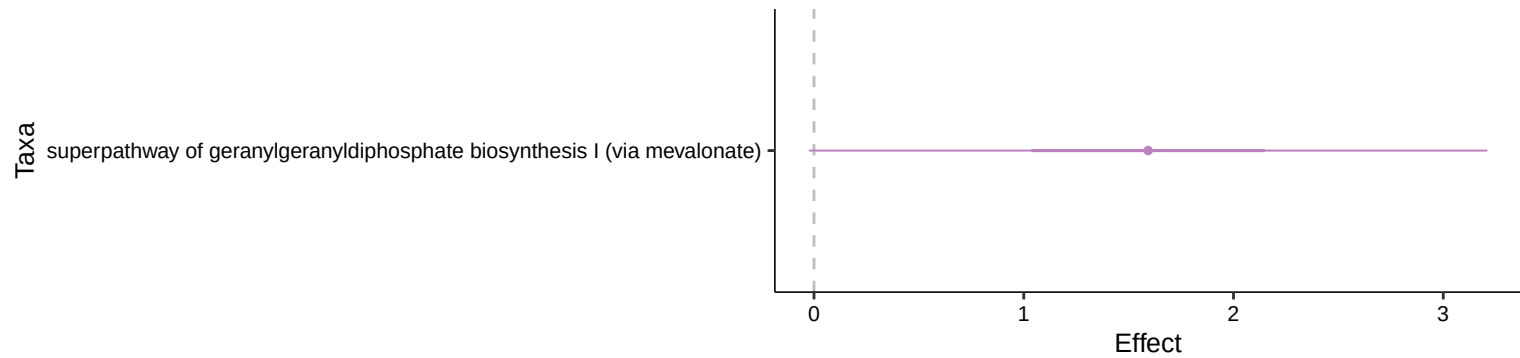
Nucleoside and Nucleotide Biosynthesis



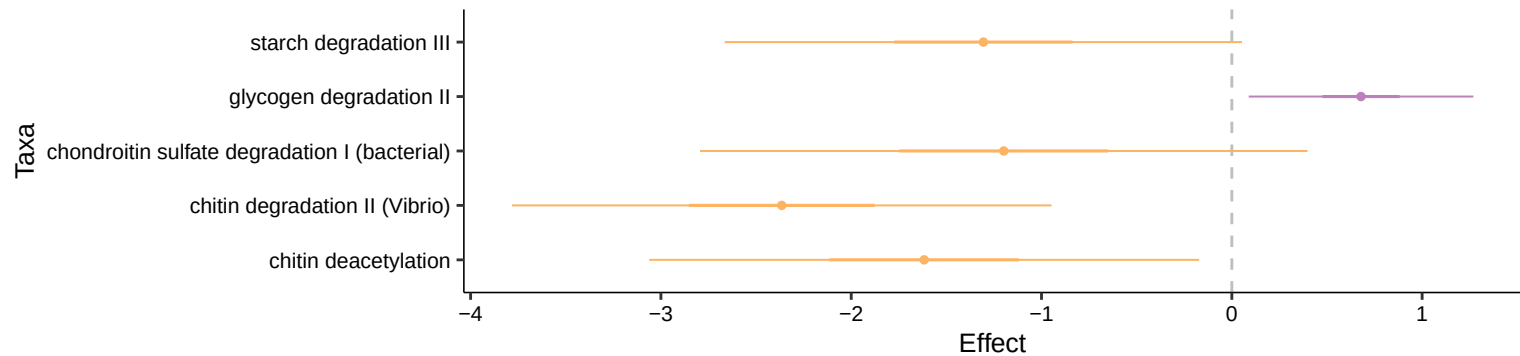
Nucleoside and Nucleotide Degradation



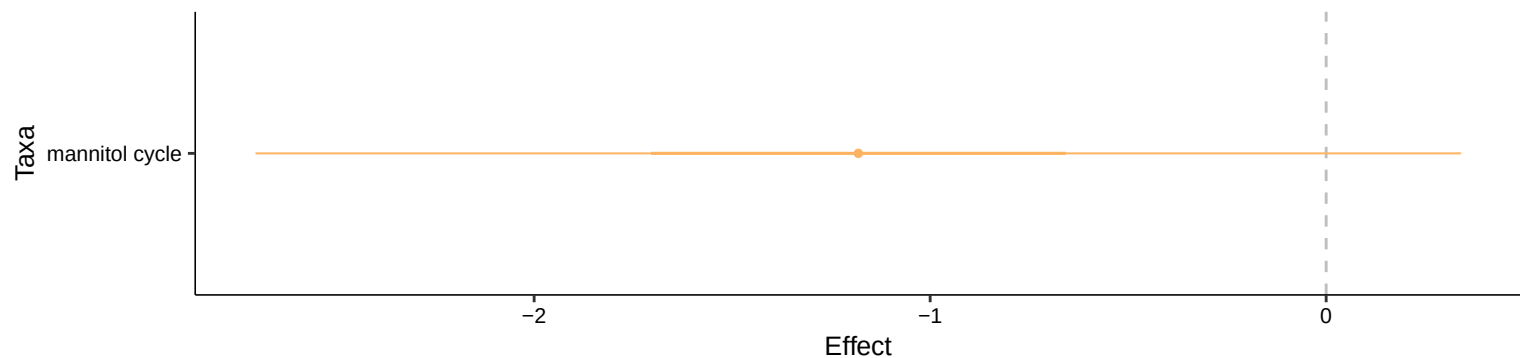
Polyprenyl Biosynthesis&&Secondary Metabolite Biosynthesis



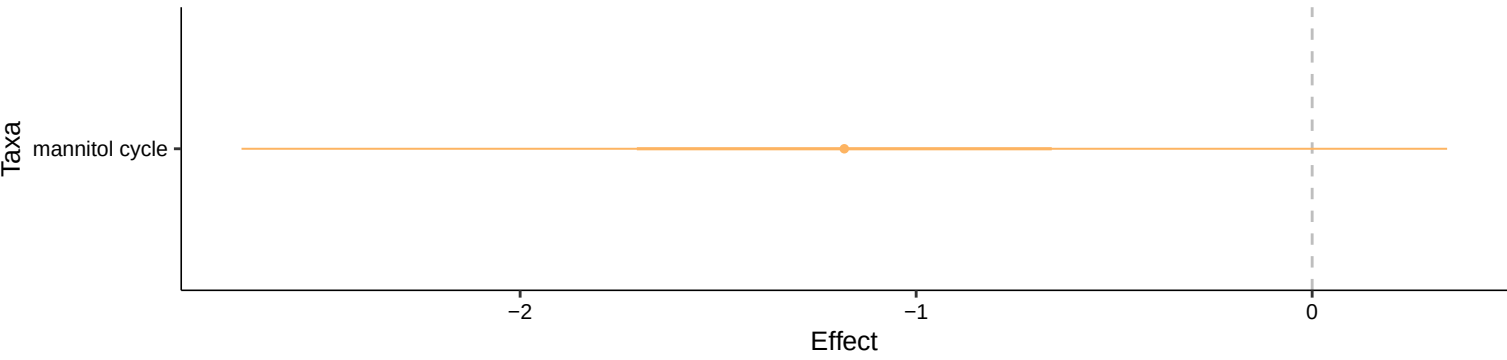
Carbohydrate Degradation&&Polymeric Compound Degradation



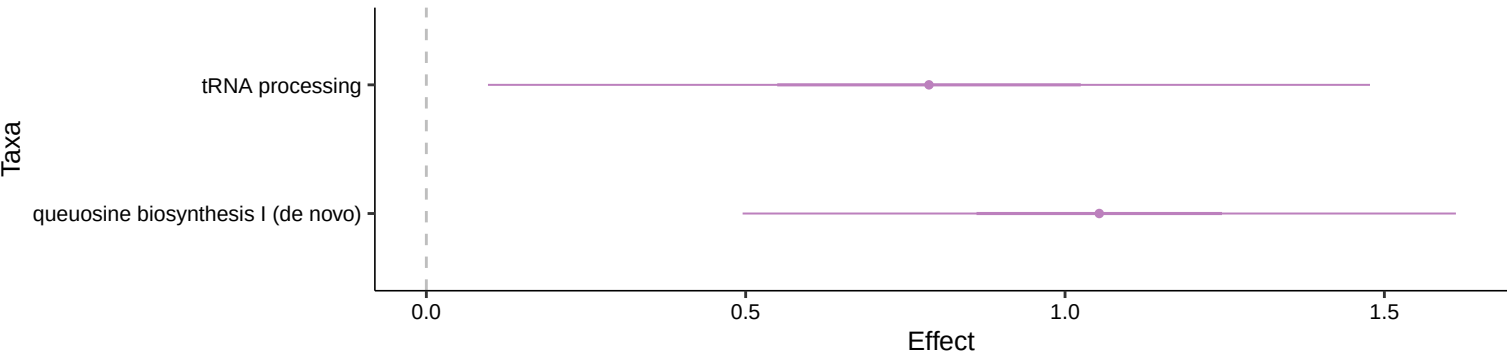
Alcohol Degradation&&Carbohydrate Degradation



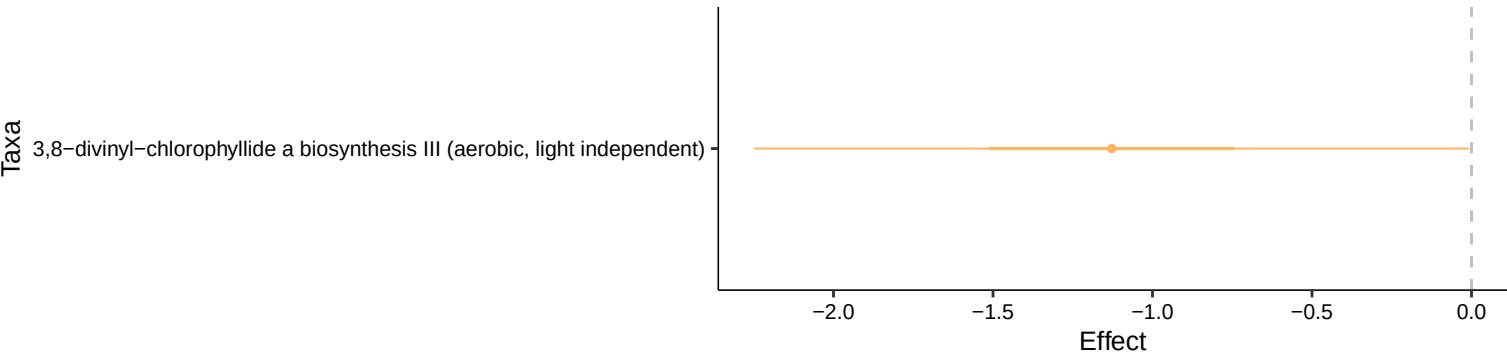
Alcohol Degradation&&Carbohydrate Degradation



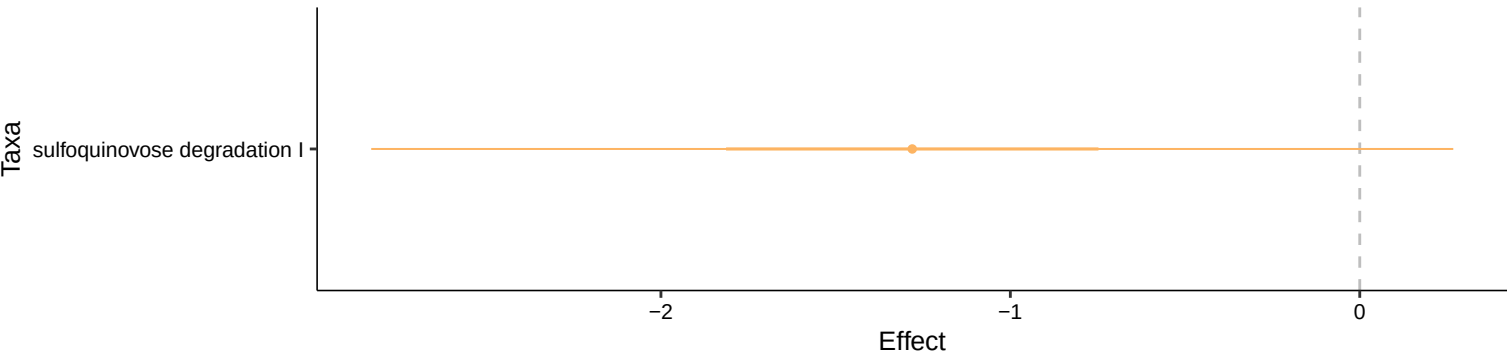
Nucleic Acid Processing



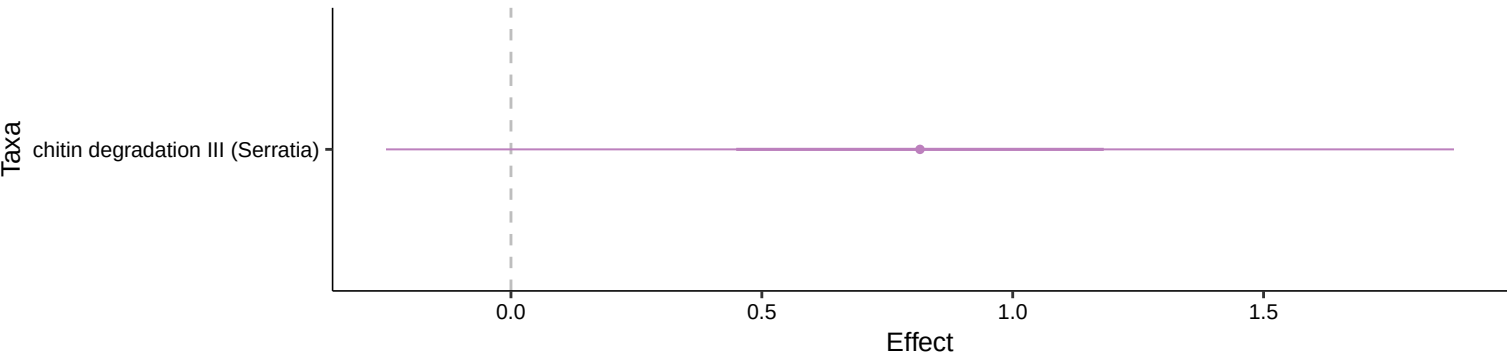
Tetrapyrrole Biosynthesis&&Secondary Metabolite Biosynthesis



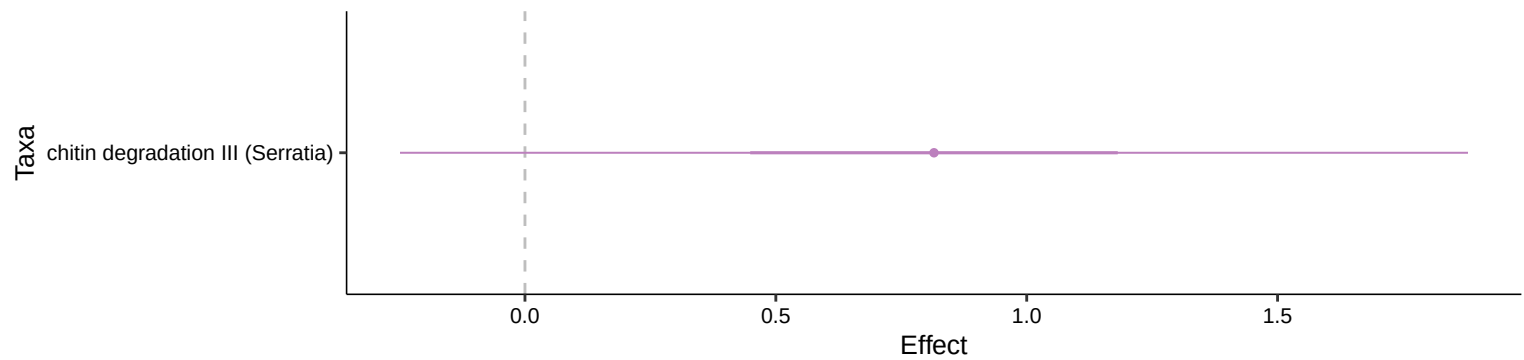
Secondary Metabolite Degradation



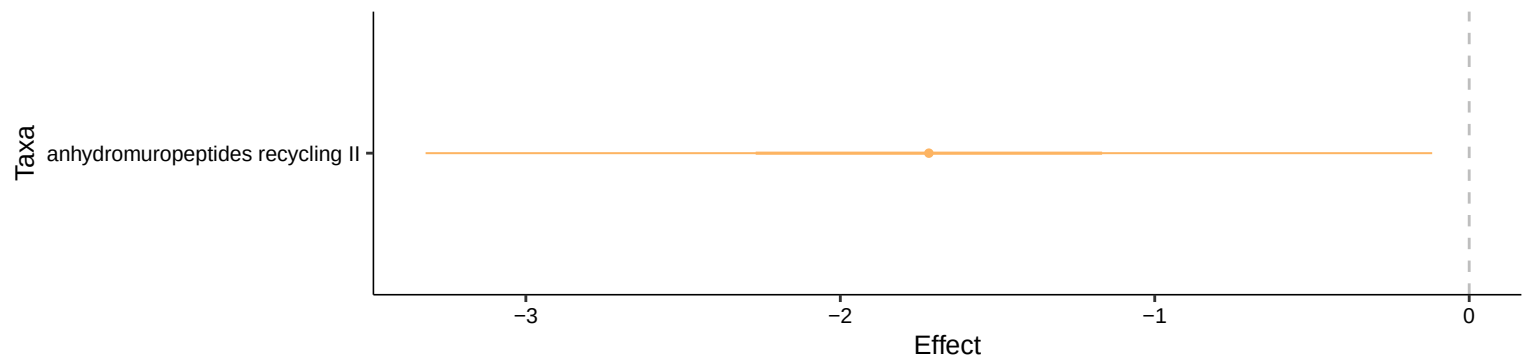
Polymeric Compound Degradation&&Carbohydrate Degradation



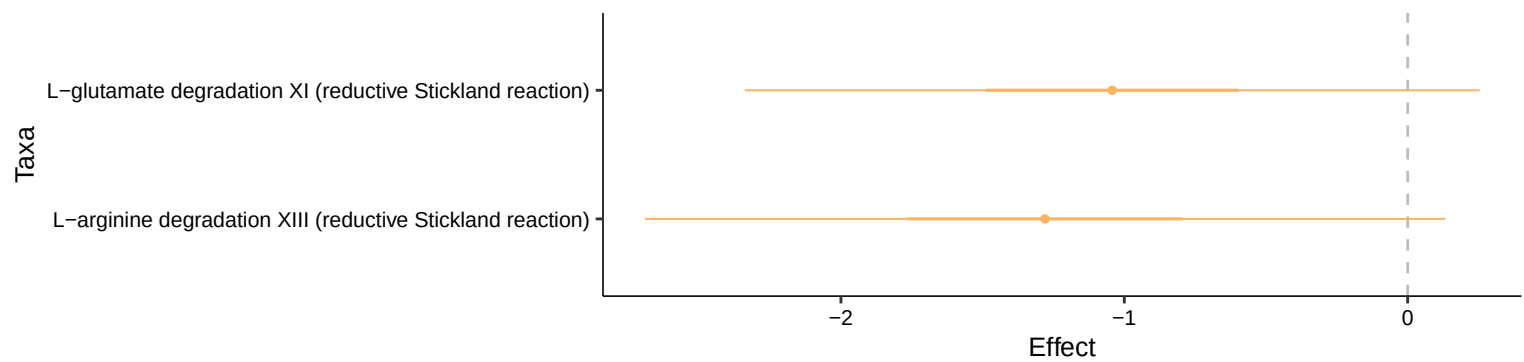
Polymeric Compound Degradation&&Carbohydrate Degradation



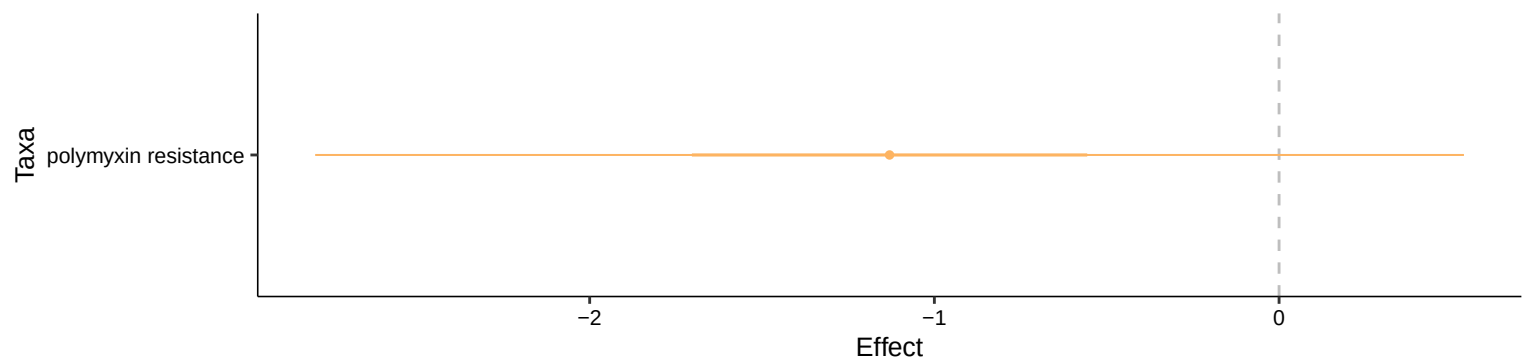
Degradation/Utilization/Assimilation – Other



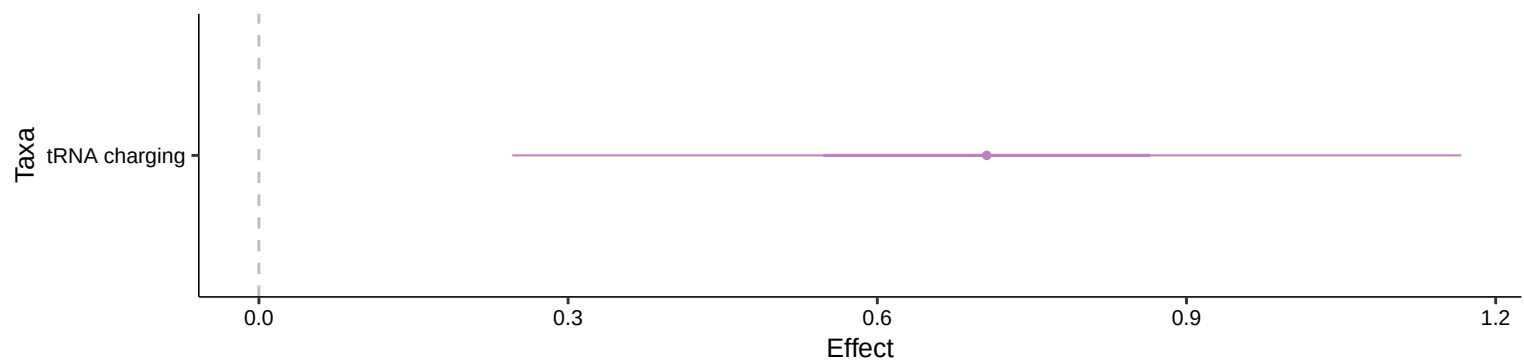
Fermentation&&Amino Acid Degradation



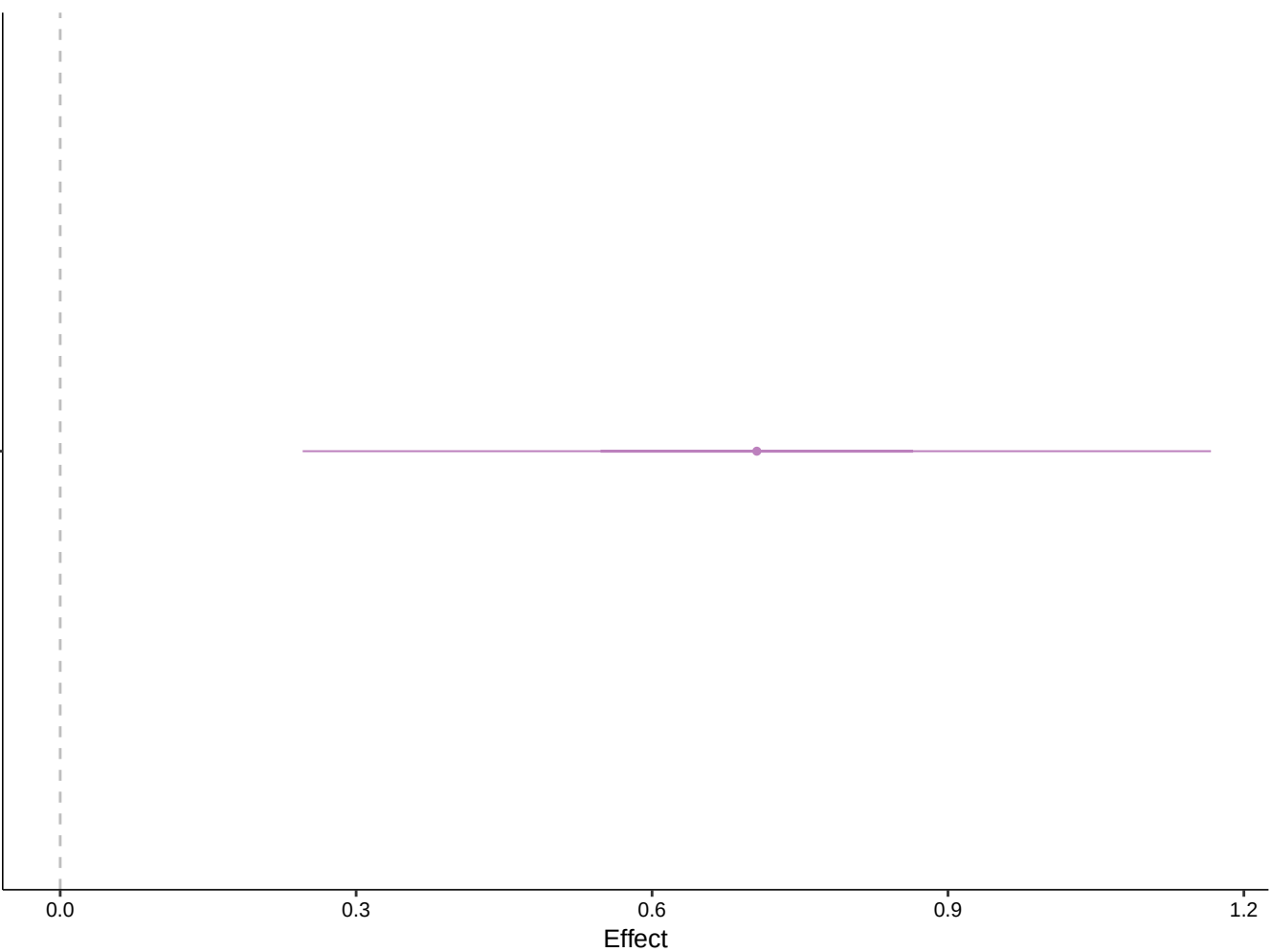
Cell Structure Biosynthesis&&Antibiotic Resistance



Aminoacyl–tRNA Charging



Aminoacyl-tRNA Charging



Amide, Amidine, Amine, and Polyamine Biosynthesis

